Low Acrylonitrile NBR

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Abstract

The patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber) reveals a focus on emulsion and solution polymerization processes. The major synthesis approaches involve the use of various emulsifiers, initiators, and chain transfer agents to control the polymerization reaction. Recurring chemistry includes the use of acrylonitrile content between 18-46 wt% and the employment of di-isopropylbenzene hydroperoxide as an initiator. Trends in the patent landscape indicate a preference for cold process discontinuous emulsion polymerization and the use of nitrogen replacement and cold water cooling to control the reaction temperature. The resulting polymers exhibit a range of properties, including Mooney viscosity and tensile strength. The patent landscape also highlights the importance of temperature control, with reaction temperatures ranging from 5-30 ■ and 20-60 ■ for emulsion and solution polymerization, respectively.

Methodology

Patent 1

- Patent Number: Not disclosed
- Patent Title: Not disclosed
- Assignee: Not disclosed
- Publication Year: Not disclosed
- Jurisdiction: Not disclosed
- Polymerization Process: emulsion polymerization
- Acrylonitrile Content: 18-46 wt%
- Monomer Ratio: Not disclosed
- Water Amount: Not disclosed
- Emulsifier: RA rosin acid or/and soap
- Initiator: di-isopropylbenzene hydroperoxide, iron edta sodium salt or secondary reduction agent rongalite
- Chain Transfer Agent: tert-dodecyl mercaptan, uncle's ten carbon mercaptan, uncle's 14 carbon mercaptan, uncle's 16 carbon mercaptan
- Temperature: 5-30 ■
- Reaction Time: 7-12 hours

- Conversion: 65-80%
- Coagulation: Not disclosed
- Polymer Properties: Mooney viscosity: 30-95, tensile strength: 10-22 Mpa
- Experimental Notes: cold process discontinuous emulsion polymerization, nitrogen replacement, cold water cooling, temperature-reducing coil

Patent 2

- Patent Number: US318330
- Patent Title: SOLUTION POLYMERIZATION OF ACRYLONITRILE
- Assignee: Not disclosed
- Publication Year: 1952
- Jurisdiction: US
- Polymerization Process: solution polymerization
- Acrylonitrile Content: at least 70 mole percent
- Monomer Ratio: Not disclosed
- Water Amount: Not disclosed
- Emulsifier: Not disclosed
- Initiator: 0.1 to 2.0% by weight based on the weight of the monomers of a free-radical-liberating catalyst
- Chain Transfer Agent: Not disclosed
- Temperature: between about 20 and 60 C
- Reaction Time: 1 to 20 hours
- Conversion: Not disclosed
- Coagulation: Not disclosed
- Polymer Properties: intrinsic viscosity of 1.8 and 2.6
- Experimental Notes: carbon dioxide dissolved in the polymerization medium to exclude oxygen

Cross-Patent Comparison & Synthesis Trends

Emulsifier Selection and Loading

The use of emulsifiers in Low Acrylonitrile NBR synthesis is evident, with Patent 1 employing RA rosin acid or/and soap. However, Patent 2 does not disclose the use of an emulsifier.

Initiator and Chain Transfer Agent Strategies

The initiators used in the patents include di-isopropylbenzene hydroperoxide and iron edta sodium salt (Patent 1) and a free-radical-liberating catalyst (Patent 2). Chain transfer agents, such as tert-dodecyl mercaptan, are used in Patent 1 to control the molecular weight of the polymer.

Monomer Ratio and Reaction Control

The monomer ratio is not disclosed in either patent, but the acrylonitrile content is specified as 18-46 wt% (Patent 1) and at least 70 mole percent (Patent 2).

Coagulation and Post-Treatment Conditions

The coagulation conditions are not disclosed in either patent, but Patent 1 mentions the use of a cold process discontinuous emulsion polymerization and nitrogen replacement.

References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year
Not disclosed	Not disclosed	Not disclosed	Not disclosed	Not disclosed
US318330	SOLUTION POLYMERIZATION OF ACRYLONITRILE	Not disclosed	US	1952